

IMU-Assisted Adaptive ECG Filtering

A Reproduction, Critique, and Improvement Study

Internship Project Report - Anastasia Lora

Reproduction target

Ou, Z., Ma, S., Zhao, Z., & Wang, H. (2025). *A Study on NLMS-Based Adaptive ECG Filtering Algorithm Assisted by Dual-Channel IMU Signals*. ISAIMS 2025, Wuhan, China. DOI: 10.1145/3777577.3777619.

1. Summary

This document summarizes my internship work on the IMU-assisted adaptive ECG filtering paper by Ou et al. (2025). I performed a strict reproduction of the paper's NLMS pipeline on the public PhysioNet PTT-PPG dataset (22 subjects $\times$ 3 activities, 500 Hz, finger IMU, 3-lead chest ECG), added an empirical critique with statistical tests and ablations and attempted to transplant the same pipeline to PPG. The work is organized that way, to be generalized as possible and point out the difficulties I faced and the mistakes I might have made, that someone could detect.

Three concrete answers that the work delivers.

- (1) *Does the NLMS work?* Yes, for baseline drift removal; 16–19 dB BPR across all activities, with the NLMS itself doing essentially all of that work (the 0.16–40 Hz bandpass contributes ~ 0 dB).
- (2) *Does the quaternion reference help?* Not in the way it is described. A literal constant $q \equiv 1.0$ yields equal or better BPR than the real quaternion across all activities (Wilcoxon value $p < 0.001$).
- (3) *Single vs. dual input?* Dual wins on BPR but the gain is not motion specific. The paper's claimed +10 percentage points R-peak retention improvement does not reproduce. The actual improvement is +0.02 to +0.15 pp, roughly 100 $\times$ smaller than the claim.

Important Disclaimer. The PhysioNet dataset places the IMU on the finger and ECG on the chest, whereas the paper's setup uses a co-located wearable.

Results worth keeping. A direct transplant of the same pipeline to PPG (pleth_2) did not produce useful denoising. The reason is structural, not a bug: the paper's preprocessing and NLMS hyperparameters are tuned for the ECG baseline band (0–0.5 Hz). It is difficult to isolate the PPG pulse because motion artifacts occupy the same bandwidth (0.5–5 Hz). We would need different frequency ranges for pre-processing the PPG signal.

2. Context of the Internship Work

Wearable ECG monitoring is degraded by two main noise sources: low-frequency baseline drift (electrode–skin impedance changes, breathing) and motion artifacts (body movement causing electrode displacement). The paper study proposes an NLMS adaptive filter that uses two IMU reference signals - integrated X-axis velocity and the real part of a quaternion to suppress both simultaneously for drift and motion artifact.

The goal had two directions. First, to reproduce the paper on a public dataset and assess how robust the reported numbers are when the original private dataset is not available. Second, to identify whether the architectural choices generalize, and if not, to propose a corrected/fixed method.

2.1 Comparison of the datasets Table

Parameter	Paper (Ou et al.)	Our setup (PhysioNet PTT-PPG)
Subjects	10	22
Activities	8 (incl. deep breathing, squat, run, bend)	3 (sit, walk, run)
Access	Private	Public (PhysioNet)
Sampling rate	250 Hz	500 Hz
IMU placement	Not explicitly stated (shown in Fig.1)	Finger clip
ECG	Single-lead wearable	3-lead chest

The IMU/ECG placement mismatch is the single most important disclaimer in this work. I mention it again in every section where it could affect interpretation.

2.2 Implementation deviations from the paper

Two equations in the paper are typographically broken and were interpreted before implementing:

- **Equation (4)** writes the small-angle quaternion update with ω_x three times in place of $(\omega_x, \omega_y, \omega_z)$. A literal implementation would ignore Y and Z rotations. Interpreted as an obvious typo.
- **Equations (7) and (8)** are identical, both referring to v . The second was interpreted to use q_{ref} (the quaternion reference).
- **Unit conversion:** gyroscope samples converted from deg/s to rad/s before the small-angle update, otherwise the approximation in eq. (4) is off by a factor of ~ 57 .
- **NLMS hyperparameters:** $\mu = 0.05$, $\epsilon = 1e-6$ (not specified in the paper; chosen for stability across all 66 records).
- **ECG standardization:** Computing the gain and baseline per subject for every .hea file, using the inversed WFDB, because of an unclear phrasing on the dataset.

3. What I Built

3.1 Strict reproduction (src/reproduction/)

An implementation of the paper's pipeline: per-subject ECG calibration from the WFDB .hea headers, 0.16–40 Hz bandpass + 50 Hz notch preprocessing, trapezoidal-rule velocity from X-axis acceleration, small-angle quaternion update with strict normalization, quaternion multiplication and extracting the real part only of quaternion, cross-correlation alignment of each reference against the preprocessed ECG, and a single-tap ($L = 1$) NLMS with two reference channels.

Key scripts:

- `benchmark_paper_strict.py` — main script. Outputs per-record BPR, MSE_{BL}, PCC, retention, Δ RR to `results/benchmark_results_paper_strict.csv`.
- `ablation_quaternion.py` — the decisive ablation: replaces the real quaternion with three surrogates (constant 1.0, linear ramp, random slow noise) to test whether motion information matters.
- `diagnostic_peak_fp.py` — proper TP/FP/FN analysis against the dataset's annotated R-peaks, with NeuroKit2 as detector and ± 50 ms tolerance.
- `statistical_tests.py` — Wilcoxon paired tests, single-vs-dual, per metric per activity.

3.2 DSP-corrected and placement-robust variant (src/improved/)

The improved method keeps the paper's two-reference NLMS architecture (linear velocity + real-part quaternion) but introduces four refinements.

(a) Weighted multi-axis velocity. The paper states that the velocity reference is the axis with the highest combined Pearson correlation and Normalized Mutual Information against the ECG noise residual. The strict reproduction uses only the X-axis (the paper's stated default) for cross-subject consistency. The improved method computes all three axis velocities (v_x , v_y , v_z) by trapezoidal integration of the centered and bandpass-filtered accelerations, computes MI and Pearson correlation, min-max normalises each metric to $[0, 1]$ and averages them into a per-axis weight. Then, uses the weighted sum $w_x \cdot v_x + w_y \cdot v_y + w_z \cdot v_z$ as the velocity reference. This way it adapts the reference to the actual motion content of each record.

(b) Bandpass-prefiltered cross-correlation for delay estimation. The reproduction follows the paper's description and cross-correlates the raw references against s_{obs} . The improved method bandpasses both the references and the target to 1–10 Hz before the cross-correlation. The idea is that the wideband references are dominated by their own slow trends and the wideband s_{obs} is dominated by QRS spikes, neither of which is a stable feature for delay alignment; restricting both to a shared motion band makes the lag estimate more robust. The maximum-correlation lag is searched in a ± 150 -sample window (± 0.3 s at 500 Hz), both for v and q .

(c) $L = 256$ -tap NLMS instead of $L = 1$. The paper specifies a memory-less NLMS (one weight per reference, applied instantly). Motion artefacts propagate through the body and electrode interface over tens to hundreds of milliseconds. A memory-less filter cannot model this. The improved method uses an $L = 256$ -tap FIR section per reference, giving the adaptation the dynamic range to track a time motion-to-electrode

transfer rather than only its instantaneous gain. The choice of $L = 256$ was set by a systematic sweep (could have tested more values). A tested sweep over $L = [1, 16, 64, 128, 256]$ and $MU = [0.05, 0.1, 0.3, 0.5]$.

(d) Quaternion reference. The real-part-of-quaternion reference is kept exactly as the paper defines it, despite the ablation in §4.2 showing it can be matched by a literal constant. The §4.2 finding is an empirical observation in the specific finger-IMU dataset and the geometric explanation between the total angular path and the θ_{net} , it is only a hypothesis that has not been validated further on other placements.

3.3 PPG transplant — negative result (src/ppg/)

I also applied the paper's exact pipeline to the PPG channel of the dataset (pleth_2 channel, co-located with the IMU). This was a 'cleanest possible' test of the paper's claims on PPG, since any motion artifact in PPG should correlate well with the IMU. The result was that the pipeline did not produce useful PPG denoising.

3.4 Analysis and reporting (src/analysis/)

Plots and aggregation are kept separate from the benchmarks. The script `figures_file1.py` emits two figures (time-domain pipeline stages overlay and PSD across stages). Tables 1–3 of the manuscript draft are generated by `table1_paper_strict.py`. `Scpit nlms_plot_errors.py` prints out the absolute difference between the three reference errors that are produced from the NLMS algorithm and plots them as well in order to verify the similarity in the efficiency. The `nlms_plot_diagnostic.py` plots the filtered signal.

4. Key Findings

4.1 Does the NLMS algorithm work?

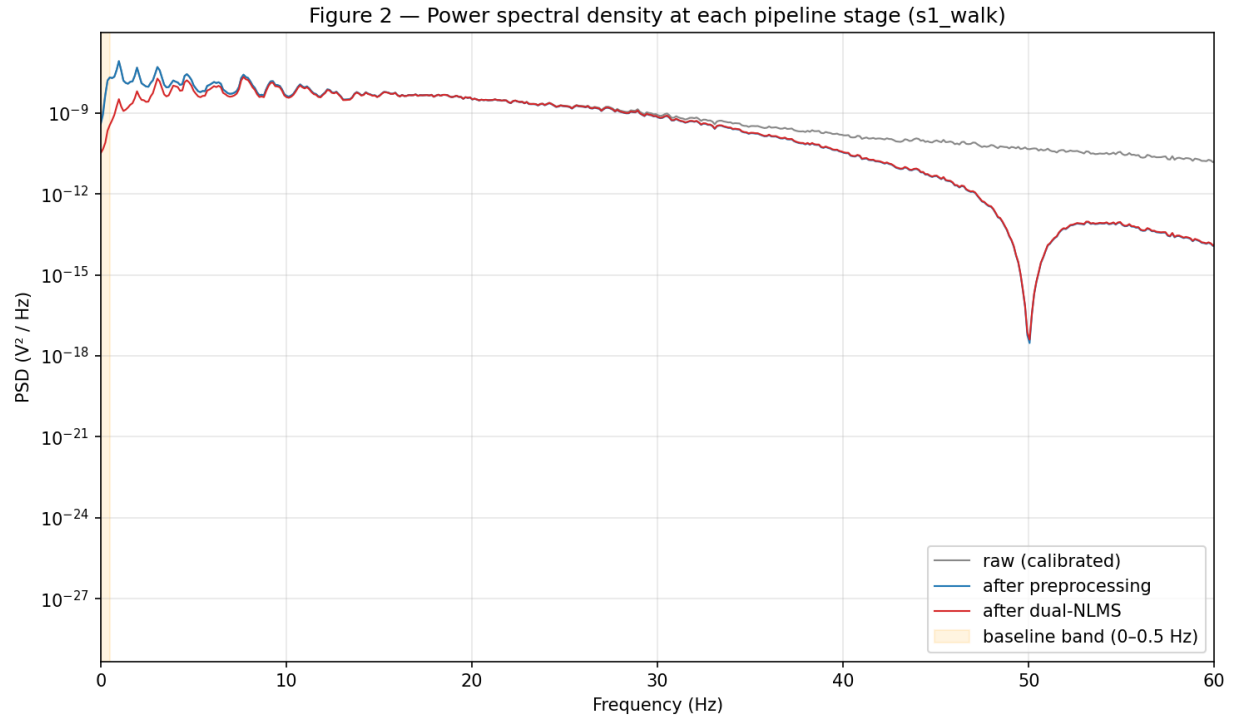
Yes, but not for the reason claimed. I decomposed the BPR into preprocessing and NLMS components, expecting that the bandpass would carry most of the load and the NLMS would not reduce by itself much baseline. BPR (Baseline Power Reduction) metric: Quantifies how much low-frequency (baseline drift) energy was suppressed by the filtering pipeline. Larger is better. Formula:

$$\text{BPR}_{\text{total}} = 10 \cdot \log_{10}(\text{P}_{\text{LF}}(\text{s}_{\text{raw}}) / \text{P}_{\text{LF}}(\text{s}_{\text{clean}})) \quad [\text{dB}]$$

Activity	BPR_preproc (dB)	BPR_NLMS only — Dual (dB)	BPR_total — Dual (dB)
Sit	-0.70 ± 3.22	17.10 ± 6.63	16.40 ± 7.66
Walk	+0.59 ± 0.57	18.37 ± 2.30	18.96 ± 2.78
Run	+0.35 ± 1.02	17.28 ± 3.64	17.64 ± 4.33

My initial hypothesis was wrong. The preprocessing contributes essentially nothing (~0 dB) and the NLMS does all the work (17–18 dB). Why: the baseline is conventionally 0–0.5 Hz, but the bandpass cuts only below 0.16 Hz. The 0.16–0.5 Hz frequency range passes through untouched, leaving the NLMS to remove it. So, the NLMS is doing real signal processing.

Visual PSD. A Power Spectral Density (PSD) plot shows how a signal’s total energy is distributed across frequencies. For an ECG signal, three regions matter: 0–0.5 Hz holds baseline drift, 5–25 Hz holds the QRS complexes (the heartbeat energy that needs to be maintained) and 50 Hz is the powerline interference spike. Figure 2 shows the PSD at three pipeline stages on a walking record: raw (gray), after the bandpass + 50 Hz notch (blue), and after the dual-input NLMS (red).



[Find in path: figures/fig2_psd.png]

Figure: PSD at three stages on a walking record. The 50 Hz powerline is removed by the notch (sharp dip in preprocessed and NLMS, absent in raw) and the content above ~40 Hz is attenuated by the bandpass. The NLMS suppression of the 0–0.5 Hz baseline band is quantified in the BPR table above - the energy removed is too narrow-banded to show as a visible separation between blue and red. (*src/analysis/figures_file1.py*).

NLMS implementation verification (the 3 error signals)

I produced three error signals $e(n)$ from the same input and reference and compared them: (i) a manual NLMS loop that was written and used throughout this project, (ii) the NLMS from the `padasip` library with the error $e(n)$ computed outside the inner iteration and (iii) the NLMS from the `padasip` library, that was faced as a reference implementation. The three signals were computed on the same record with identical μ , ϵ , and reference inputs.

All of the error signals are visually indistinguishable. The conclusion was that the manual NLMS loop that was written at the start of the internship is correct and matches the reference from the `padasip` library. Both the output of samples and the plots showed any significant difference between the two implementations.

Script: `src/analysis/nlms_plot_errors.py`.

4.2 Does the quaternion reference contribute motion information?

No. I replaced the real quaternion with three reference substitutes that contain *no* motion information: a literal constant ($q \equiv 1.0$), a linear ramp, and slow random noise. If the quaternion were doing motion-specific work, the substitutes should all underperform.

Activity (BPR, dB)	single	dual_real	dual_const	dual_ramp	dual_random
Sit	12.44	17.10	19.95	18.15	16.38
Walk	4.73	18.37	18.65	12.75	14.43
Run	5.64	17.28	18.16	11.96	13.49

The constant $q \equiv 1.0$ — a signal that, by construction, carries no motion information — **matches or beats** the real quaternion in BPR across every activity (Wilcoxon $p < 0.001$). Whether it is being mentioned ‘single’ refers to single – only accelerometer – input and ‘dual’ refers to dual – accelerometer & gyroscope – input as reference channels in the NLMS algorithm.

As a side observation, I computed an Oscillation Index — $OI = \int |\omega| dt / \theta_{\text{net}}$, the ratio of angular path length to net rotation, as a geometric diagnostic to explain why a constant might be matching the quaternion. The numbers ($OI \approx 110$ in walking and $OI \approx 88$ in running) suggest that the quaternion real part discards most of the actual angular motion of the IMU, but I did not pursue this further.

4.3 Single-input vs. dual-input NLMS

Dual wins on BPR, but the gain is not motion-specific. Dual is better (the difference is small) than single in BPR/MSE_{BL}/R_{BL} — but adding a constant in the second channel produces equal or better numbers. So the right answer would be: a second tap in the NLMS helps, but it helps because the filter gains a second degree of freedom for DC tracking, not because the IMU's rotational state matters.

The paper's headline gain is the +10 percentage points R-peak retention improvement of dual over single. I tested this with proper TP/FP/FN against the dataset's annotated R-peaks on all 66 records:

Activity	Signal	Sens %	PPV %	FP / record
Sit (n=22)	s_obs	99.79	99.57	2.32
	s_clean_single	99.77	99.54	2.46
	s_clean_dual	99.79	99.52	2.64
Walk (n=22)	s_obs	99.39	99.27	5.09
	s_clean_single	99.28	99.13	6.05
	s_clean_dual	99.40	99.30	4.91
Run (n=22)	s_obs	99.35	99.35	4.68

Activity	Signal	Sens %	PPV %	FP / record
	s_clean_single	99.33	99.27	5.27
	s_clean_dual	99.38	99.38	4.46

Dual-vs-single improvement is +0.12 pp in walking, +0.05 pp in running, and +0.02 pp in sitting. The paper's +10 pp claim does not appear to be reproducible on this dataset: the actual best-case gain is roughly 100× smaller.

4.4 Methodological issues with the paper's metrics

PCC reference is not specified

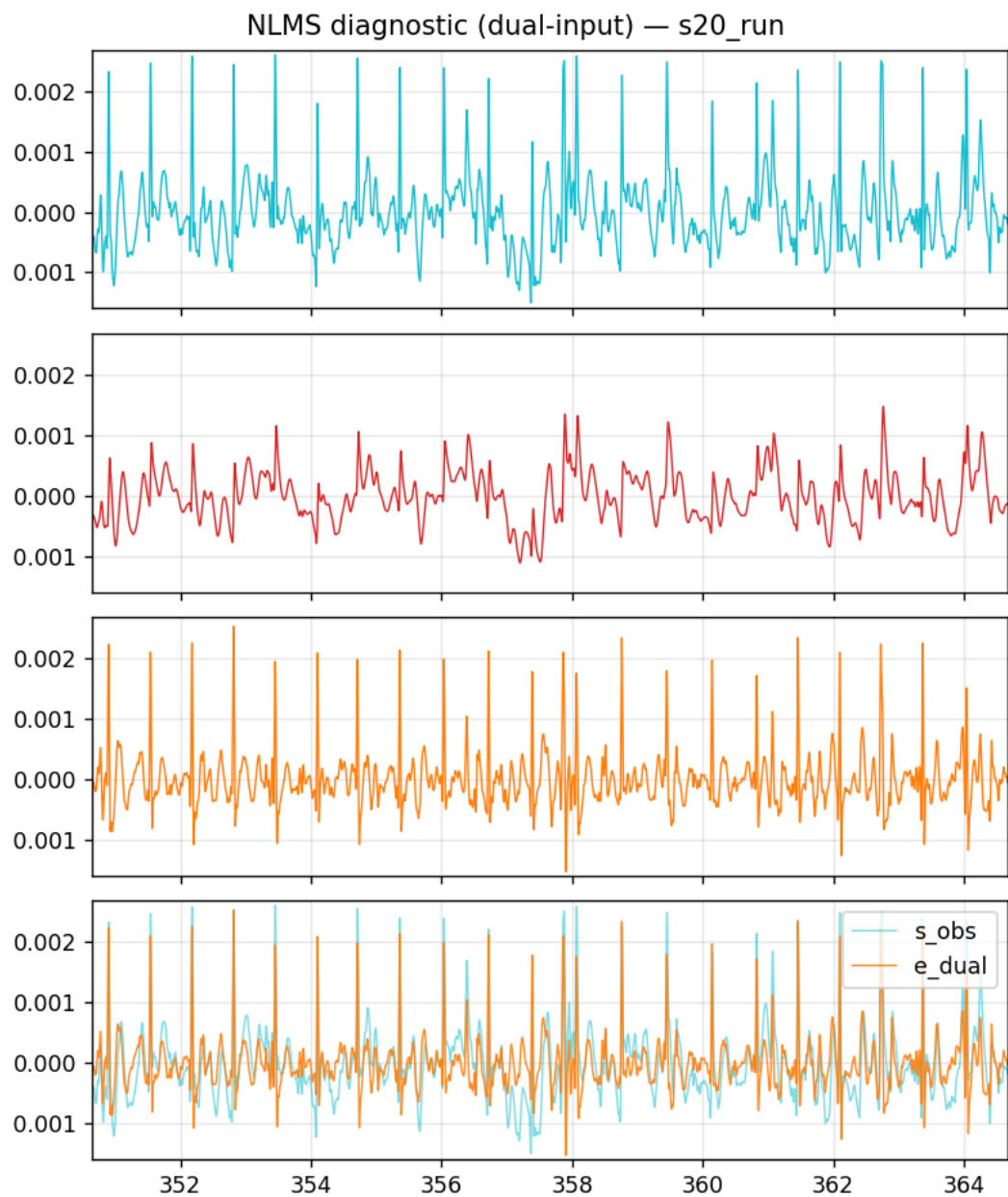
The paper computes a Pearson correlation between the cleaned ECG and ‘the clean reference’, but never specifies what ‘clean’ means. A) If the raw signal was going to be used (`s_raw`) — then a perfect filter would drive PCC to 0, which is going against papers’ claims. B) If `s_obs` (the preprocessed signal) was used — then a high PCC means the NLMS did nothing. C) ground-truth clean ECG — ideal but unavailable in any real wearable dataset. In this implementation it was used B) and obtained PCC = 0.77–0.81 versus the paper's 0.91–0.93.

Retention rate is ceiling-bound and ignores false positives

When ECG is reasonably clean, retention rate hits 99–100% on every method, so it cannot discriminate. It also counts true positives only, ignoring false positives. A filter that adds false peaks scores the same as one that does not. This is the main reason the paper's claimed +10 pp single→dual gain does not surface in our 66-record run: the metric saturates well below the visible threshold. The analysis I ran per-record TP/FP/FN computation against the dataset's ground-truth R-peaks can be found in `src/reproduction/diagnostic_peak_fp.py`.

BPR conflates preprocessing and NLMS

The paper reports a single 33.98 dB BPR for the full pipeline and attributes it to the NLMS. This number is not being reproduced. Decomposing BPR (§4.1) shows the bandpass contributes ~0 dB and the NLMS contributes the full ~17–18 dB. It is calculated for clarification and specification reasons, in order to understand the metric itself.



[Find in path: analysis/python nlms_plot_diagnostic.py -h]

Figure: ECG figure showing plots starting from top to bottom; a) s_{obs} : The preprocessed Filter, b) $y = W * x$: The noise that NLMS removed, c) e_{single} : The cleared signal output and d) an Overlay between s_{obs} and e_{dual} .

4.5 Statistical significance — paired Wilcoxon tests

All the single-vs-dual comparisons above are point estimates averaged across 22 records per activity. To check whether the per-record differences are statistically meaningful and not just averaging artifacts, I ran a paired Wilcoxon signed-rank test for each metric and activity.

The script `src/reproduction/statistical_tests.py` outputs both Wilcoxon and paired t-test p-values.

Metric	Sit (n=22)	Walk (n=22)	Run (n=22)	Verdict (across activities)
BPR (total)	p<0.001 ***	p<0.001 ***	p<0.001 ***	Dual better
BPR (NLMS only)	p<0.001 ***	p<0.001 ***	p<0.001 ***	Dual better
MSE_BL	p=0.002 **	p<0.001 ***	p<0.001 ***	Dual better
R_BL	p=0.003 **	p<0.001 ***	p<0.001 ***	Dual better
PCC (vs s_obs)	p<0.001 *** (dual worse)	p<0.001 ***	p<0.001 ***	Mixed (dual worse in sit)
Retention	p=1.000 ns	p=0.037 * (dual worse)	p=0.110 ns	No reproducible gain
Δ RR (ms)	p=0.011 *	p<0.001 ***	p<0.001 ***	Dual better

On BPR, MSE_BL, R_BL, and Δ RR — the metrics that respond to baseline-drift removal — dual is significantly better than single across activities ($p < 0.001$ for walk and run and weaker p values in sit, which is expected because there is less baseline content to remove when the subject is stationary). On Retention Rate the metric the paper relies on for its +10 pp claim, but there is no significant single-vs-dual difference in sit and run, and a small statistically significant *dual-worse* result in walk, combined with the ceiling-bound caveat.

5. What Didn't Work

5.1 PPG transplant

PPG implementation. Same exact pipeline as the paper (bandpass 0.16–40 Hz, 50 Hz notch, integrated X-axis velocity, real-part quaternion, single- and dual-input NLMS), but with the target signal swapped from ECG to pleth_2 (IMU-co-located PPG channel). The expectation was that PPG would be the cleanest test of the paper's claims, since here the motion source and the IMU are physically the same body part.

The pipeline runs end-to-end without errors but the 'denoised' PPG is, qualitatively, no cleaner than the input. The diagnostic plots from `src/ppg/nlms_plot_diagnostic_ppg.py` show that the NLMS error signal contains roughly the same motion as the signal that went through the pre-process (`s_obs`: bandpass + notch filtering).

Quantitative HR validation. Using the WFDB-annotated R-peaks from the simultaneously-recorded ECG as ground truth (no detector uncertainty on the truth side), I compared HR derived from PPG — raw vs. NLMS-cleaned, single- and dual-input — in 8-second sliding windows. The validation script is `src/analysis/metrics_cross_modality.py`. Two metrics: mean absolute HR error (MAE, lower is better) and the IEEE 1708a-2019 coverage threshold (Cov5, percentage of windows with $|\text{error}| < 5$ bpm).

Activity	N	MAE_raw (bpm)	MAE_dual (bpm)	Cov5_raw (%)	Cov5_dual (%)
Sit	22	0.65	15.63	99.0	57.2
Walk	22	8.56	19.26	65.6	29.8
Run	22	3.27	14.22	84.0	43.8
All	66	4.13	16.36	83.0	43.6

The NLMS fails to improve PPG-derived HR. In sit, where the raw PPG already gives essentially clinical-grade HR (MAE = 0.65 bpm, 99% coverage), the dual NLMS pushes MAE up by a factor of 24 (to 15.63 bpm) and collapses coverage from 99% to 57%. Over all 66 records, MAE quadruples (4.13 to 16.36 bpm) and coverage drops from 83% to 43.6%. The transplanted pipeline is not just neutral on PPG, it is harmful.

NLMS energy ratio across activities. The diagnostic script `src/ppg/nlms_plot_diagnostic_ppg.py` reports, for any chosen window, the fraction of input signal energy that the NLMS removes: $\|y\|^2 / \|s_{\text{obs}}\|^2$ (where $y = W \cdot x$ is the component the filter estimates and subtracts). The script's own interpretation note flags >50% as an over-subtraction regime. At that level the filter is removing far more than the motion artifact and is necessarily eating into cardiac content. The threshold for an ideal filter on a sitting subject (no motion to subtract) is <5%. Three records of evidence, one per activity (100-second windows) are summarized below:

Record	Activity	single (%)	dual (%)	Window (s)
s18_sit	Sit	81.1	83.4	100–200

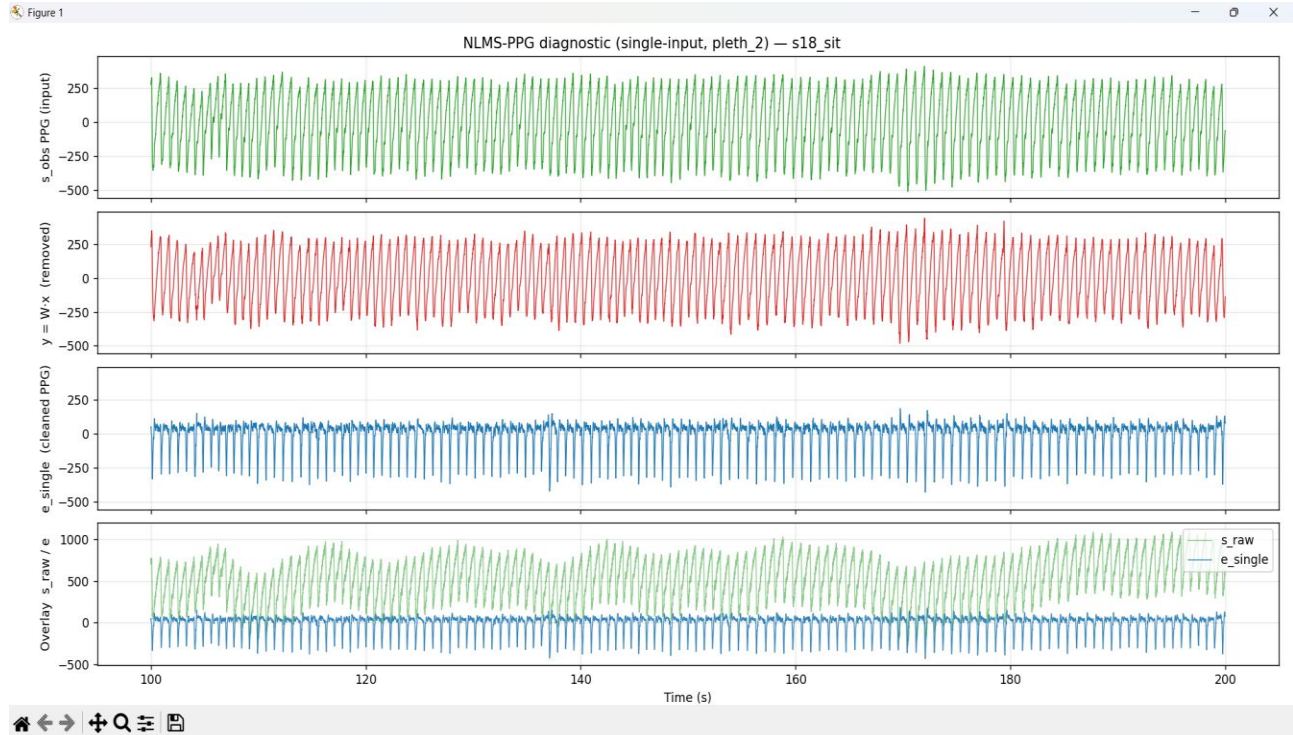
s11_walk	Walk	100.5	93.6	100–200
s13_run	Run	103.4	89.7	100–200

Activity **Sit**: The subject is stationary and there is no motion artifact for the filter to remove, yet the NLMS removes **83.4% of the input energy**.

Activity **Run**: The single-input ratio exceeds 100% (103.4%). This is mathematically admissible — NLMS minimizes $\|e\|^2$ without bounding $\|y\|^2$, so when the references are well-aligned with a periodic component of s_{obs} the adaptation can overshoot locally. What it indicates is that the integrated-velocity reference v is strongly correlated with the heartbeat itself, after the cross-correlation lag step locks it onto the pulsatile rhythm.

This is the mechanistic explanation for the HR-MAE collapse in the table above. The NLMS is not failing to clean the signal — it is locking onto the pulsatile signal via reference leakage and subtracting it. More information about the diagnostic figures can be found at the end of this section. The y panel (what the filter removes) is a near-perfect copy of the s_{obs} panel (the input) — the cleaned output e is essentially the residue left after the heartbeat has been subtracted.

PPG vs ECG. The paper's preprocessing was designed around the ECG noise structure: baseline drift in 0–0.5 Hz, QRS in 5–25 Hz, large spectral separation. The NLMS adapts on slow correlation between IMU references and the target, then subtracts. In PPG, the useful pulsatile signal sits in 0.5–5 Hz, which is also where motion artifacts live. The bandpass does not separate signal from artifact in PPG, and the NLMS adaptation cannot distinguish them either — attempting to remove motion in that band attenuates the heartbeat. A second confirming detail comes from the BPR decomposition: on ECG (§4.1) preprocessing contributes essentially 0 dB and the NLMS does the full 17–18 dB of baseline-band suppression. On PPG, preprocessing alone already produces 18–21 dB of BPR — the 0.16 Hz high-pass cutoff is removing a large fraction of low-frequency PPG content (DC drift, slow trends) before the NLMS ever runs. The NLMS then adds another 9–20 dB on top, and as the energy ratio above shows, much of that removal lands inside the pulsatile band.



[Find in path: figures/ppg_figure_X_input_RECORD-FILE.png], where X refers to single or dual input and RECORD-FILE refers to the printed record.

Figure: PPG channel before and after the (transplanted) NLMS (produced by `src/ppg/nlms_plot_diagnostic_ppg.py`).

In conclusion, direct transplantation of the paper's pipeline to PPG is not viable. A PPG-specific pipeline would need (a) a PPG-tuned bandpass (typically 0.5–8 Hz), (b) a reformulation of the noise model that does not assume artifact and signal are spectrally separable, and (c) likely a different filter such as wavelet decomposition rather than NLMS subtraction.

5.2 Initial BPR-decomposition hypothesis was wrong

I expected the preprocessing bandpass to do most of the baseline drift removal, with the NLMS contributing marginal additional suppression. The decomposition (§4.1) showed the opposite. The mistake was assuming the bandpass cut-off (0.16 Hz) was inside the baseline band, but the bulk of the baseline energy (0.16–0.5 Hz) passes through the filter untouched, leaving the NLMS to perform the primary suppression.

5.3 Co-located IMU experiment was not possible

The strongest test of the paper's claims would require an IMU placed near the ECG electrodes (e.g., chest-patch) rather than on the finger. I could not find any public dataset that would meet this requirement. This is a dataset limitation, not an implementation problem.

Appendix A — Repository File Map

Path	Role
README.md	Repo navigation, setup, how-to-run.
src/reproduction/benchmark_paper_strict.py	Strict implementation of the paper's pipeline. Outputs per-record metrics to results/.
src/reproduction/ablation_quaternion.py	Replaces the real quaternion with three substitutes.
src/reproduction/diagnostic_quaternion.py	Gyroscope + quaternion statistics across the dataset (the Oscillation Index data).
src/reproduction/diagnostic_peak_fp.py	TP/FP/FN analysis against dataset-annotated R-peaks, NeuroKit2 detector.
src/reproduction/statistical_tests.py	Wilcoxon paired tests across all single-vs-dual comparisons.
src/improved/benchmark_improved.py	New file with improvements.
src/improved/ NLMS_parameters_test.py	Sweeps parameters L over [1, 16, 64, 128, 256] and mu over [0.05, 0.1, 0.3, 0.5]
src/ppg/benchmark_paper_strict_ppg.py	Paper pipeline transplanted to PPG (pleth_2 IR).
src/ppg/ablation_quaternion_ppg.py	PPG version of the quaternion ablation.
src/ppg/nlms_plot_diagnostic_ppg.py	PPG diagnostic plots for the filtered . final signal.
src/analysis/figures_file1.py	Generates figures/fig1, fig2, fig3 (time domain, PSD, OI bars).
src/analysis/table1_paper.py	Aggregates per-record CSV into Table 1 (mean $\pm$ std by activity) for ECG.
src/analysis/table2_paper_ppg.py	Aggregates per-record CSV into Table (mean $\pm$ std by activity) for PPG.
src/analysis/metrics_cross_modality.py	Cross-modality (ECG/PPG) consistency checks, making use of MAE metric.
src/analysis/nlms_plot_errors.py	Diagnostic plots for the NLMS error signals and weight evolutions.
src/analysis/nlms_plot_diagnostic.py	ECG diagnostic plots for the filtered/ final signal.
results/*.csv	All per-record metric outputs.
figures/*.png	Generated plots.

Appendix B — Reproducibility

Reproduction order:

```
python src/reproduction/benchmark_paper_strict.py
python src/analysis/table1_paper.py
python src/reproduction/ablation_quaternion.py
python src/reproduction/diagnostic_quaternion.py
python src/reproduction/diagnostic_peak_fp.py -full
python src/analysis/nlms_plot_errors.py -h
python src/analysis/nlms_plot_diagnostic.py -h
python src/reproduction/statistical_tests.py
python src/analysis/figures_file1.py
python src/ppg/benchmark_paper_strict_ppg.py
python src/analysis/table2_paper_ppg.py
python src/ppg/ablation_quaternion_ppg.py
python src/ppg/nlms_plot_diagnostic_ppg.py -h
python src/analysis/metrics_cross_modality.py -h
python src/improved/benchmark_improved.py
python src/improved/NLMS_parameters_test.py
```

Dataset is the public PhysioNet Pulse Transit Time PPG dataset , available at <https://physionet.org/content/pulse-transit-time-ppg/>. Place the unpacked directory at Physionet_PTT_Dataset/ in the project root before running.